Journal of Cleaner Production 252 (2020) 119801

Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Electronic waste collection systems using Internet of Things (IoT): Household electronic waste management in Malaysia

Kai Dean Kang [a], Harnyi Kang [a], I.M.S.K. Ilankoon [a, *], Chun Yong Chong [b]

[a] Discipline of Chemical Engineering, School of Engineering, Monash University Malaysia, Jalan Lagoon Selatan, Bandar Sunway, Selangor Darul Ehsan, 47500, Malaysia
[b] School of Information Technology, Monash University Malaysia, Jalan Lagoon Selatan, Bandar Sunway, Selangor Darul Ehsan, 47500, Malaysia

ARTICLE INFO

Article history:
Received 2 September 2019
Received in revised form
13 December 2019
Accepted 18 December 2019
Available online 21 December 2019

Handling Editor: Cecilia Maria Villas Bôas de Almeida

Keywords:
E-Waste/electronic waste
Mobile apps
Smart solid waste systems
Recycling in Malaysia
Smart cities
Sustainable development goals (SDGs)

ABSTRACT

Waste electric and electronic equipment or e-waste generation has been identified as a key aspect in solid waste management, though e-waste disposal in landfills is not suggested due to the toxic chemicals and heavy metals in it. The presence of valuable metals, namely precious and base metals, such as gold and copper, respectively also highlights the importance of effective waste management strategies. Even though some developed countries practice modern household e-waste management techniques, including extended producer responsibility (EPR) schemes, household e-waste legislative frameworks are not yet fully implemented in Malaysia. In order to support the sustainability of smarter cities concept, household e-waste needs to be efficiently managed and this work discusses the application of smart collection systems pertaining to the Malaysian e-waste management and recycling sector. A smart household e-waste collection box was designed, fitted with e-waste level measurement sensors to record the disposal data. A backend server was developed which automatically notifies and schedules e-waste collectors to dispatch and collect the e-waste when the volume of the collection box reaches a certain threshold (e.g. box is 80% filled). A mobile application was developed in this work and public end users are intended to use it to dispose their household e-waste. The smart system was successfully developed as a proof-of-concept in this work and it could be beneficial to improve the household waste consumer electronics collection in Malaysia.

© 2019 Elsevier Ltd. All rights reserved.

1. Introduction

Electronic waste or e-waste generation is considered to be one of the fastest growing solid waste streams in the world (Ilankoon et al., 2018) and it contains 10 different categories (e.g. large and small household appliances, consumer equipment, lighting equipment) based on Annex 1A, the EU waste electrical and electronic equipment directive (UNEP, 2007a, b). It is envisaged that the global e-waste generation is 44.7 million tonnes in 2016 (Baldé et al., 2017). Southeast Asian countries and China have greatly contributed to the generated e-waste volume and the value in Asia in 2016 is 40.7% of the global e-waste generation (Baldé et al., 2017). There are a few reasons for these trends; firstly, the reduced replacement interval of electrical and electronic equipment (EEE) contributes to generate more e-waste. Secondly, the proliferation of consumer EEE (e.g. Printed circuit boards or PCBs manufacturing growth in Asia, Huang et al., 2009) has increased the generated e-waste volume (Baldé et al., 2017). In addition, e-waste management, especially household e-waste, has not been efficiently employed in Asia and this may be one of the reasons why e-waste generation in Asia has started to grow significantly as recycling operations will not be effective without established e-waste management frameworks (Ilankoon et al., 2018).

It is not recommended to dispose e-waste in landfills (Greenpeace International, 2009) due to its hazardous nature (Spalvins et al., 2008). It typically contains noxious metal containing components and chemicals (Sepúlveda et al., 2010), for example, halogenated flame retardants present in non-metal fraction (Cobbing, 2008). If e-waste were disposed in landfills, long term metal leaching would occur and the leachates containing toxic metals and chemicals above certain concentrations in soil and water could cause environmental hazards and human health risks. This implies the requirement of safe disposal of e-waste by managing this solid waste stream efficiently (Song and Li, 2015) and it

* Corresponding author.
E-mail address: saman.ilankoon@monash.edu (I.M.S.K. Ilankoon).

https://doi.org/10.1016/j.jclepro.2019.119801
0959-6526/© 2019 Elsevier Ltd. All rights reserved.

has recently gained strong interest in solid waste management sector (Ilankoon et al., 2018).

One of the effective ways of managing the e-waste streams has been e-waste reuse, recycling, and value recovery operations. Reuse and recycling operations are more sustainable (Nowakowski, 2018) and those generate refurbished consumer electronic equipment and functional electronic components (Ilankoon et al., 2018), especially in developing countries (Yong et al., 2019). Value recovery techniques are feasible in the industrial context due to high metal concentrations (Fernando et al., 2019), such as gold and copper, in e-waste compared to the same in respective metal ores (Liang et al., 2010). This aspect drives project economics in e-waste value recovery operations. Thus, e-waste is considered a secondary metal resource (Yong et al., 2019) and value recovery operations are known as urban mining techniques (Van Eygen et al., 2016). The generalised e-waste composition was presented by Widmer et al. (2005) and Ari (2016) and it consists of 60% metals, 15% plastics, 5% metal-plastic mixture, and 2% PCBs. PCBs comprise 30% metals (copper is the most concentrated metal) (Kumar et al., 2014), while the non-metal fraction consists of resins, plastics and ceramics (Kim et al., 2004). In a study in Belgium, it was found that value recovery operations save 80 and 87% of materials in the case of desktops and laptops, respectively (Van Eygen et al., 2016). However, recycling of the plastic fraction has received limited attention in previous research studies (e.g. Guo et al., 2009) and considered to be an important research gap considering the overall e-waste volume (Ilankoon et al., 2018).

It is imperative to employ efficient e-waste management systems in order to carry out successful e-waste reuse, recycling, and value recovery operations. Baldé et al. (2017) estimated that only 20% (8.9 million tonnes) of the e-waste generated in 2016 is documented to be collected and properly recycled. It shows the inefficiencies of e-waste management systems and this situation is very significant for household e-waste (e.g. Malaysia) and e-waste generated by industries is typically governed by regulations and legislative frameworks enacted by the governments.

In a recent article published in the Journal of Cleaner Production, Yong et al. (2019) studied the current status of e-waste management and value recovery operations in Malaysia in detail. The Environmental Quality Regulations 2005 in Malaysia categorise e-waste as a scheduled waste, SW110 (DOE, 2005) and it is differentiated as industrial and household e-waste. Currently, industrial e-waste in Malaysia is handled by the legislative frameworks and an online portal, namely, electronic scheduled waste information system (eSWIS) is employed by the industries to record e-waste management data (eSWIS, 2019). However, household e-waste is not currently governed by any legislative frameworks. The department of environment (DOE), Malaysia (JICA, 2014), and other governmental and non-governmental bodies employed some initiatives (e.g. collection boxes, discarded mobile phone collection programs) to manage household waste EEE in the country (Yong et al., 2019). However, it has been difficult to predict the generation (Jayaraman et al., 2019) and material flows for household e-waste (Yong et al., 2019) and thus it was identified that collection mechanisms of household e-waste need to be improved to manage the household e-waste generation in the country (Yong et al., 2019). In order to address this research gap, the suitability of mobile applications to improve existing household e-waste management systems was suggested by Yong et al. (2019) and that will be tested in the current work. It is supposed that smart e-waste management systems improve the collection of household e-waste fraction in Malaysia, however, those have not been explicitly discussed in previous e-waste management and value recovery research studies. Since smart systems are mainly applied in conventional solid waste management (section 2), the objective of this paper is to employ smart household e-waste management systems pertaining to the Malaysia e-waste management. None of the previous household waste EEE collection initiatives in Malaysia employed smart systems and only the conventional collection boxes were used to collect e-waste items, such as mobile phones, laptops, and cables. The design of a smart collection box with mobile applications (i.e. consumer side or e-waste disposal) and data server systems (i.e. e-waste collector side/e-waste recycling company) is thus justified in the Malaysian household e-waste management sector.

2. Overview of smart waste management systems

Within the smart city concept smart systems have been designed and employed in conventional solid waste management (Nižetić et al., 2019) to keep and maintain cleaner cities (e.g. Singapore). Smart waste collection systems using internet of things (IoT) facilitate improved collection routes (i.e. lower transportation costs) and material sorting (Alcayaga et al., 2019), and ensure sustainable product life cycle management (Liu et al., 2019). These systems consist of a sensor system, microcontrollers (as a low-powered computational unit), and global system for mobile communication (GSM), such as Wireless Fidelity (Wi-Fi) module. The employed hardware components were photoelectric , radio frequency identification (RFID), weight, infrared and ultrasonic sensors. The sensors must be carefully designed to avoid drawbacks, such as overlap of the area of detection causing false detection and infrared sensor errors due to reflected sunlight (Papalambrou et al., 2015). The typical software were cloud servers to store data and mobile/web applications as user interfaces (Yusof et al., 2018), which can provide easy guidance and important information to the users (Satyamanikanta and Narayanan, 2017). The users can scan the attached RFID tags and dispose the waste and the other sensors (e.g. weight sensor) record the required information for the authorities (Satyamanikanta and Narayanan, 2017). Samann (2017) and Yusof et al. (2018) presented an idea of having solar cells or rechargeable batteries to power the components of the smart solid waste collection systems. In addition, Samann (2017) explored the possibility of implementing sleep mode on GSM module to reduce power consumption. Omar et al. (2016) proposed the use of spatial data of the collection boxes to distribute the responsibilities and optimise the solid waste collection. The level of the waste bin was accurately obtained using a Matlab simulation (Papalambrou et al., 2015). However, Yusof et al. (2018) recommended a two-way authentication method using the collection box level indicator and collection box short messaging service (SMS) notification indicator to alert the decision makers and solid waste collection company.

In other economies, Chen et al. (2018) discussed IoT based energy management framework for machine workshops to improve energy efficiency and alleviate energy costs by shutting off unnecessary tools and auxiliary equipment via real time energy aware scheduling. In medical applications, Catarinucci et al. (2015) developed a smart hospital system to collect real time data on both environmental conditions and patients' physiological parameters using RFID. An advanced monitoring application was established based on the data in a centralised server. Al-Ali et al. (2017) tagged home appliances with data acquisition modules to collect energy consumption data. The data stored in a centralised server was used to build a smart energy management system fostering smart homes concept. In addition, Minoli et al. (2017) discussed IoT applications for smart building with energy optimisation and Sodhro et al. (2019) proposed IoT frameworks to build smart cities.

3. Smart e-waste collection system architecture

3.1. Overview of system architecture

Fig. 1 depicts the overall architecture of the proposed system. A mobile application was developed to guide the users (i.e. people who dispose their e-waste) to the nearest e-waste collection box, based on their current global positioning system (GPS) location. Smart e-waste collection boxes are installed at designated locations. The collection boxes are equipped with microcontrollers with ultrasonic sensing and wireless capabilities in order to exchange information with a centralised cloud database server. In the administrative server, data are retrieved from the cloud database server in order to calculate the level of e-waste (i.e. volume of disposed e-waste) at each collection box. If a certain threshold has achieved (e.g. collection box is 80% filled), an automated email will be sent to the collection company (e.g. Meriahtek (M) Sdn Bhd in Malaysia). The company can then send the e-waste collection truck to collect e-waste from the relevant boxes.

3.2. Smart e-waste collection box

The existing design of the e-waste collection box installed by Meriahtek (M) Sdn Bhd at Monash University, Malaysia is shown in Fig. 2a, while Fig. 2b illustrates the smart e-waste collection box. It is noticeable that the volume and the opening size of the smart collection box is smaller. The small opening prevents any unauthorised access to the e-waste deposited in the collection box. The small volume of the collection box is for testing purposes as the volume of the collection box can always be changed according to the design requirements.

In order to enable wireless connectivity and sensing capability, a microcontroller is installed on each of the collection box. The three main components, as shown in Fig. 3, are the (a) HC-SR04 ultrasonic sensor, (b) Arduino board and (c) Wi-Fi module board. The ultrasonic sensor is capable of determining the level of the collection box. The Arduino board is to supply power for the ultrasonic sensor and allow customised program to control the sensor and Wi-Fi module.

In our test environment, a portable power bank was used to power-on the Arduino board. Due to the battery-powered supply for the smart e-waste system, the program needs to be tweaked to improve battery efficiency. In best case scenario, if the Arduino board is connected to a stable power source, the ultrasonic sensor can be turned into the always-on mode (i.e. constantly calculate the level at 0–1 s interval). However, it was not the case in this work (it is not required as well) and here it mimics an energy-efficient way to operate the smart e-waste collection box. Fig. 4 illustrates the workflow of the proposed energy-efficient operation mode.

By default, the system is set to sleep mode to improve the battery efficiency. Users are required to scan the QR code on the collection box to register the location code of the box (Fig. 2b). This feature is used as a validation mechanism to ensure that the user is indeed disposing e-waste at the right collection box. Since users must scan the QR code, it can be used as the trigger to wake the Arduino board and sensor up. Whenever a registered user scans a valid QR code on the collection box, the backend server will capture this event and send a wake-up signal to the system.

The ultrasonic sensor will then operate and scan for any changes in the level of the collection box. By default, we pre-set the ultrasonic sensor to operate at an interval of 3 s and the maximum wake-up duration was set at 5 min. Based on several trial runs and it was found that a new user would typically spend less than 5 min to complete the whole process, hence the 5 min threshold was

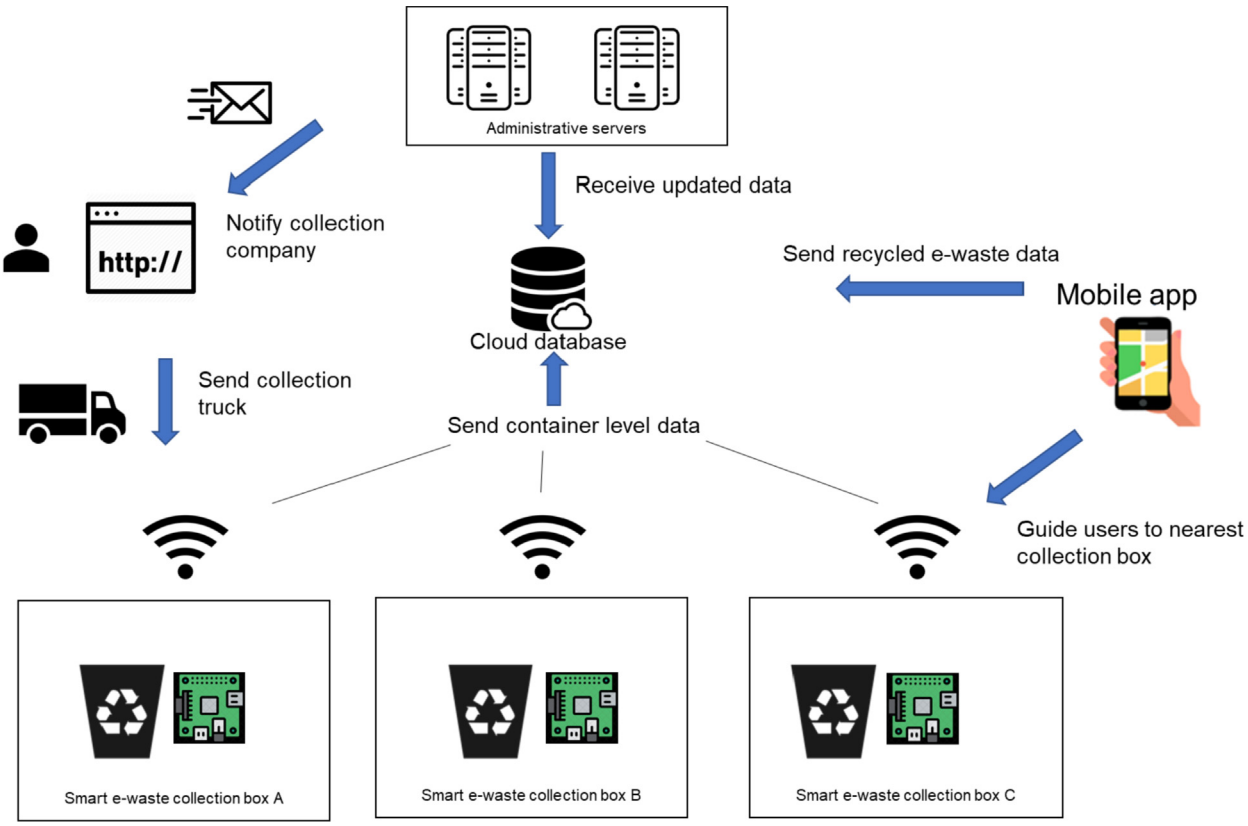


Fig. 1. Proposed smart e-waste collection system architecture.



Fig. 2. Existing e-waste collection box installed at Monash University, Malaysia campus (a) and the box designed in this work to test the smart collection system (b).

employed. If the collection box level indeed changed, it means that the users had successfully deposited the e-waste. The system will then be put into sleep mode again. If the level of the collection box did not change, the system will check if the 5 min threshold is met. If Yes, it will be turned into sleep mode again. Otherwise, it will continue to sense for changes in the collection box level.

As for the Wi-Fi module board, it serves to connect the Arduino board with the Wi-Fi network and the database server to store the level measured via the ultrasonic sensor.

The average overall current consumption of the proposed smart system during wake-up period is determined to be 80 mA (measured using a digital multimeter), though the same during the sleep mode is assumed to be 20 μA (ESP8266 Shop, 2019). Assuming that on average, there are two e-waste disposal activities every 1 h period (i.e. 10 min of wake-up period and 50 min of sleep mode), when connected to a power bank with battery capacity of 10000 mAh (battery self-discharge percentage is 15%), power bank is expected to last for 26.5 days with sleep mode operation. On the other hand, the power bank will only last for 4.4 days without the implementation of sleep mode.

In terms of the costs for the development of the smart e-waste collection system, about 37.59 USD were spent for the sensors (cost of the collection box is fixed). This smart collection system must be operated in a Wi-Fi enabled environment. It is supposed that the costs for Wi-Fi will be cheaper or free of charge due to the initiatives proposed by the Malaysian communications and multimedia commission (MCMC) to promote the transition towards smart cities (MCMC, 2019). In addition, the electricity consumption of the smart system was calculated (two e-waste disposal activities per hour) as 0.072 kWh per month and costs 3 Malaysian ringgits (0.7 USD) (Tenaga Nasional Berhad, 2019).

3.3. Cloud database server

The mobile application (Section 3.5) and Wi-Fi module board will be linked to a cloud-based database system so that the information can be collected and stored securely. Google Firebase (Fig. 5) was selected as the cloud database in this work due to its free-to-use and efficient concurrency control features. Due to its support in concurrency control, it allows system admin to effortlessly navigate and monitor all relevant data online even if there are multiple data being pushed and pulled at the same time, at multiple collection boxes.

Fig. 5 shows the console of the Google Firebase for one of the collection boxes, where (a) shows the latest level measured by the ultrasonic sensor, and (b) shows the three roots, which are the

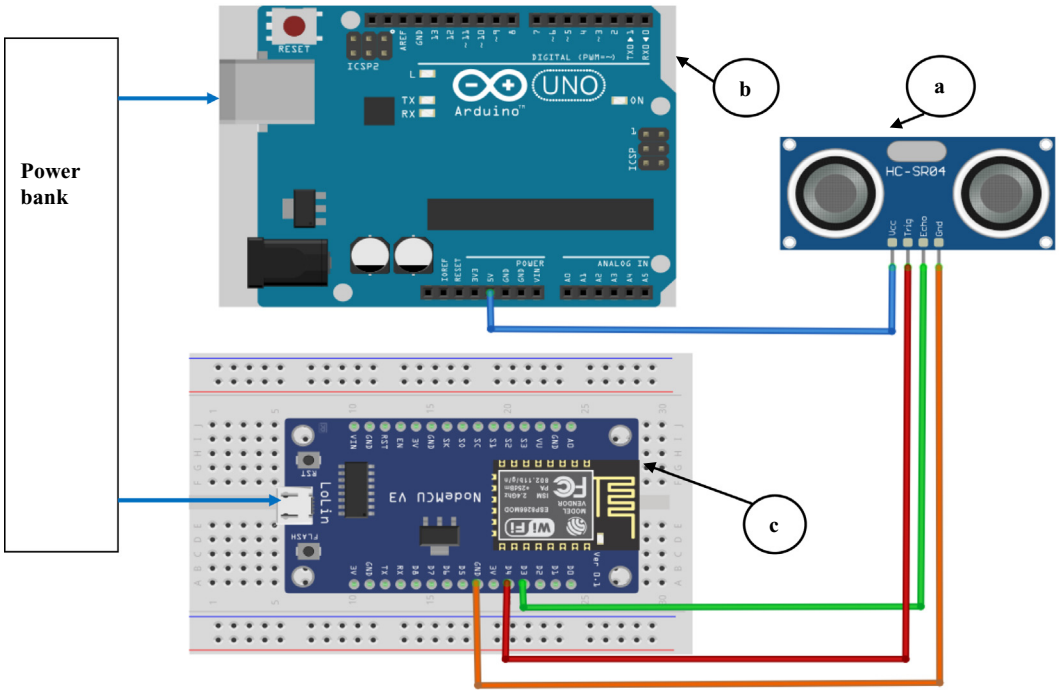


Fig. 3. Hardware used to operate the smart e-waste collection box: (a) HC-SR04 ultrasonic sensor, (b) Arduino board and (c) Wi-Fi module board.

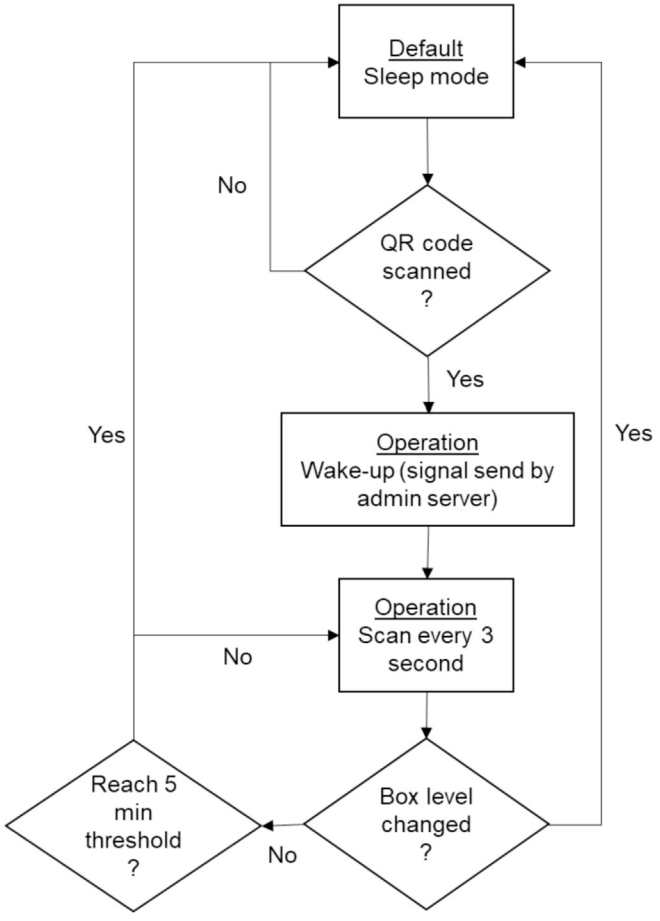


Fig. 4. General workflow of the proposed energy-efficient operation of the smart e-waste collection box.

"Level", "Submit" and "Users". Each of the roots will contain their corresponding information to ease data processing needs. The data being displayed under each of the roots are shown in Fig. 6.

In Fig. 6, the root "Level" shows the collection box e-waste level. On the other hand, "Submit" shows the image of the e-waste captured by the users, the location of the collection box, the type of disposed e-waste, unique ID for the transaction, quantity, time of the transaction, and user's ID (Section 3.5). In order to know more about a particular user, the root "Users" shows a more detailed

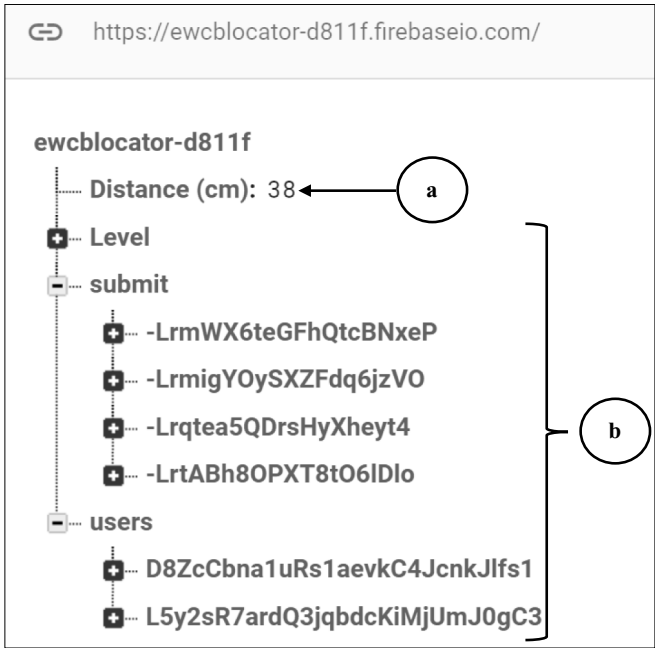


Fig. 5. Layout of database system: (a) the latest level measured by the ultrasonic sensor, and (b) the three roots, namely, "Level", "Submit" and "Users".



Fig. 6. Data displayed under each root shown in Fig. 5.

description about the registered user. Due to privacy concerns, the name and personal information about the registered users can also be encrypted.

3.4. Automated email alert system

Whenever the level of a collection box reaches a certain threshold, an automated email will be sent to the collection company. In the experimental setup, a threshold value of 80% was used (i.e. the box is 80% filled by volume), which can be further tweaked according to the needs of the e-waste collector. Utilising Gmail's simple mail transfer protocol (SMTP) server, the Arduino board on the collection box was configured to send an email to the collection company notifying them the current status of the box. The e-waste collection company can then plan their collection route accordingly based on the collected e-waste volume of different collection boxes. Since the focus of this paper is not on optimising collection routes, it is not discussed in detail. There are multiple scenarios where the collection company can choose to mobilise their collection truck. For example, collection boxes A and B are only 5 km apart. If the level of collection box A and B reaches 85% and 50%, respectively, the collection company can choose to collect from both the boxes to minimise the number of trips, even though the level at collection box B has not reached the pre-defined threshold (i.e. 80%). This aspect will be addressed in a follow up paper.

3.5. Mobile application

Figs. 7 and 8 show the proof-of-concept user interfaces. Note that the mobile app is the main interface between the users (anyone who wish to dispose of their household e-waste) and the administrative server. Login (Fig. 7a) and register (Fig. 7b) pages are set to be the launcher activity of the developed mobile application. Users are required to login in order to track their e-waste disposal activities. The developed mobile application offers the choice of navigation to the nearest collection box (Fig. 7c–e) and subsequently displays the history of the disposed items.

Google Maps is the second activity of the developed mobile apps. By utilising Google Maps application programming interface (API) and the users' GPS signal, the mobile app is capable of routing the users to the nearest e-waste collection box. The purple marker in Fig. 7c shows the current location of the user. Users can then choose the existing red markers on the Google Maps (Fig. 7c–e), which indicate the nearby e-waste collection boxes. Polylines, which indicate the direction from the current position to the destination will be drawn automatically (Fig. 7d and e). If required, users can enable the turn-by-turn navigation feature of Google Maps to reach the collection box (Fig. 7e). Additionally, a user can search for collection box location in the region by clicking the 'SEARCH' button (Fig. 7c).

At the destination, user can carry out e-waste disposal in the e-waste deposit page by clicking 'NEXT PAGE' button (Fig. 7c). At each e-waste collection box, a unique QR code is pasted on the box to act as a verification method (Fig. 2b). The users are required to scan the QR code to register the location code of the box to the mobile app. The user is supposed to take a photo of the waste EEE items before their disposal (Fig. 8a) and it will be used for subsequent validating purposes to award the reward points. In addition, users can upload already taken photos from their device photo album. E-waste deposit page entries, such as categories of e-waste (e.g. mobile phones, laptops and tablets) and quantity (i.e. number) need to be completed before clicking the 'SUBMIT' button (Fig. 8a and b). The categories of accepted e-waste items can be seen by clicking the dropdown list (this list can be modified according to the collector's preferences) as shown in Fig. 8b.

Users can access the e-waste disposal history at the history page by clicking the 'REFRESH' button (Fig. 8c). Data, such as location, disposed items, quantity, and time stamp of e-waste disposal are retrieved from the Firebase server.

In order to encourage the users to dispose their household e-waste, reward points will be awarded depending on the type, size, and quantity of the recycled items (i.e. mimics EPR schemes relevant to electronic waste and see Ilankoon et al., 2018). A proof-of-concept reward point system is shown in Fig. 8d. As such,

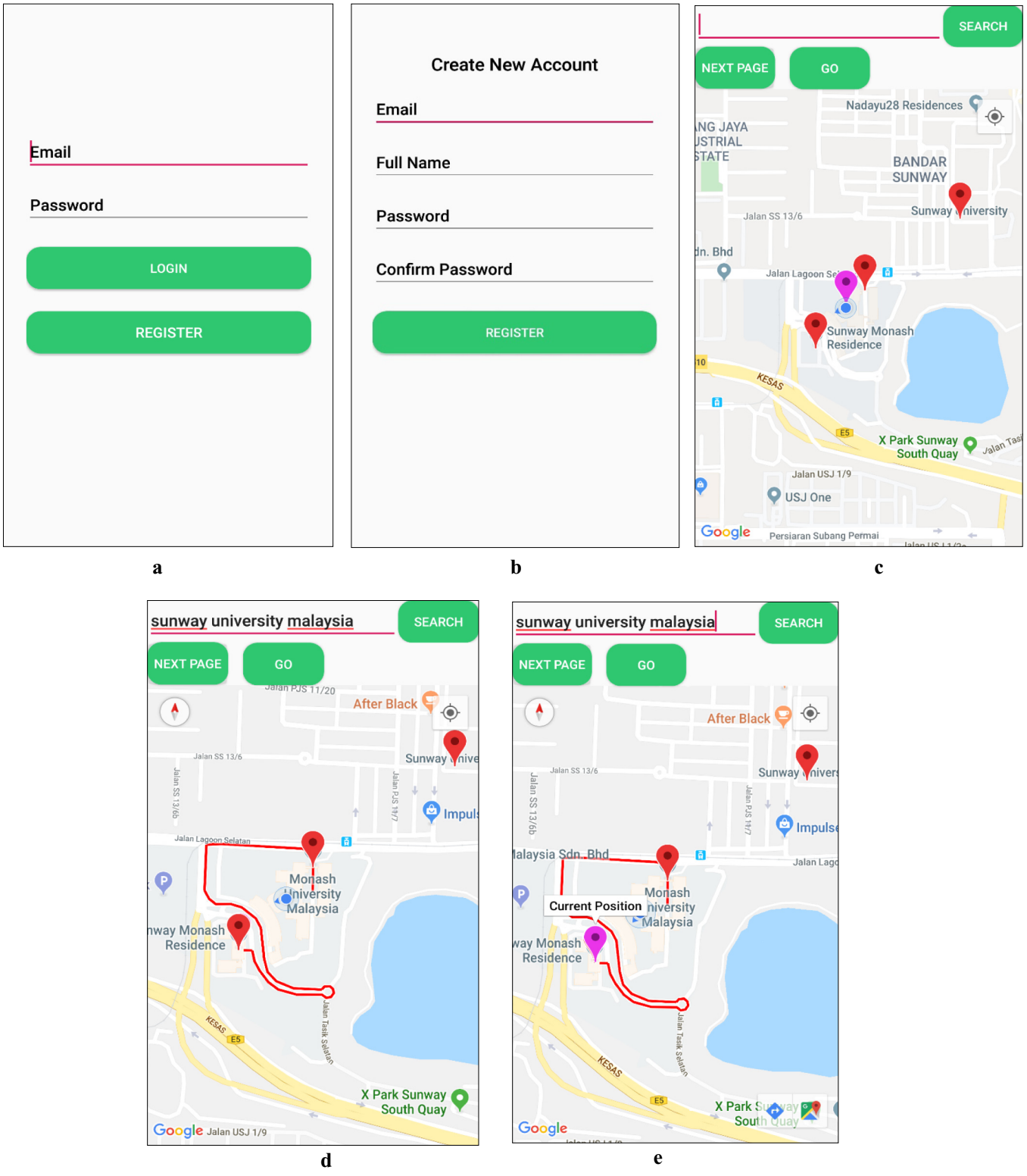


Fig. 7. User interfaces of the developed mobile application: login (a), registration (b), and user navigation (c, d and e) pages.

whenever users register to the system, a unique record is created on the centralised cloud database server. The implemented feature also ensures the persistence of data such that when users change their devices, their recycling history is not lost.

4. Testing of the smart household e-waste collection box

Fig. 9 shows the plot of the level of a collection box (measured from bottom, Fig. 2b) in the experiments when e-waste were gradually deposited into it. Different household e-waste categories,

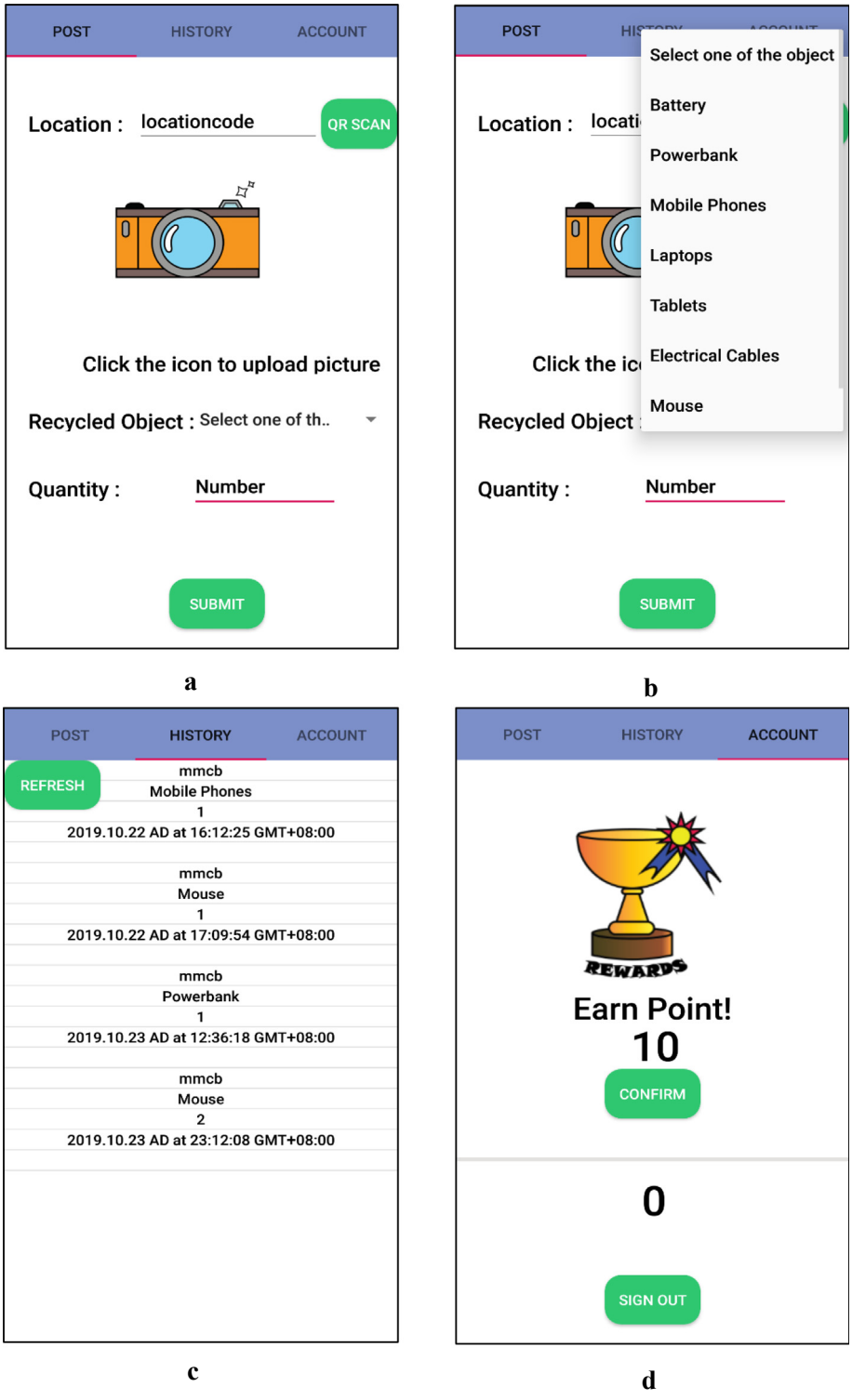


Fig. 8. User interfaces of the developed mobile application: e-waste disposal (a, b), history (c), and reward points (d) pages.

such as laptops, mobile phones, PCBs, and chargers were added during the testing stage (Fig. 2b) and the collected volume of e-waste is recorded by the level measured by ultrasonic sensor. The implemented level threshold value of 80% is also shown in the figure.

The figure generally shows the increasing level as a function of time, though some level variations (i.e. dropping the level while adding e-waste) were also observed. There are a few reasons for this trend; firstly, the displacements of e-waste in the box while adding larger and heavy pieces of e-waste (e.g. laptops) might have caused level variations at a particular time interval (e.g. 260–300 s). Secondly, the measurement errors of the ultrasonic sensor could result in random level variations (e.g. 140–180 s). This implies the necessity of sensor optimisation and it will be addressed in a follow

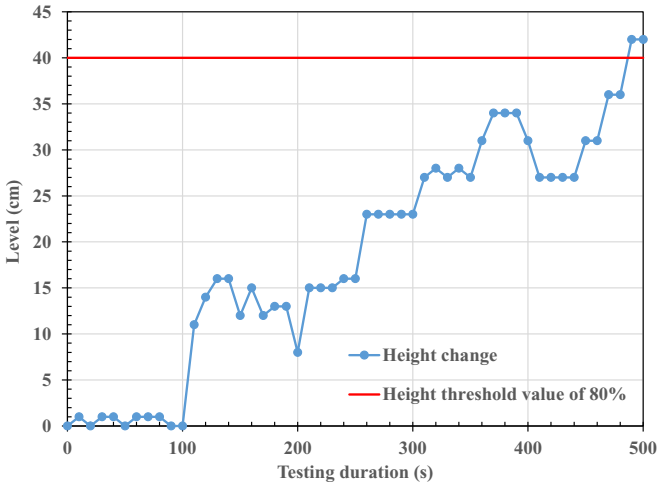


Fig. 9. Variation of the e-waste collection box level as a function of time during e-waste disposal.

up paper. Since the e-waste collector is more interested to know about the average level of the box (i.e. filled volume) to plan the e-waste collection rather than the exact volume of the collected e-waste, the level variations can be justified to a certain extent considering the practical aspects of this research (i.e. develop economic e-waste management frameworks for developing countries).

5. Possible implementation of the IoT framework in the Sunway City, Malaysia

In this section, we discuss a plausible implementation of the developed smart e-waste collection system at a large scale, using one of the sub-urban cities in Malaysia as an example. The value chain of EEE should be circular considering the "closing the loop" of materials and this implementation thus facilitates circular economy based society (Reuter and van Schaik, 2016).

Sunway City (managed by the Sunway Group) is a sub-urban city in Subang Jaya, Selangor state, Malaysia, with an estimated population of 500,000 and positions itself as a higher education hub with two international universities, namely, Monash University Malaysia and Sunway University. A local electronic waste recycling company, Meriahtek (M) Sdn Bhd, has set up a number of conventional household e-waste collection boxes (Fig. 2a) at these two universities as part of their corporate social responsibility (CSR) program (Meriahtek, 2018). The main rationale behind this decision is university students tend to carry EEE with them for learning and recreational purposes, such as smart phones, laptops, portable batteries, and removal drives. Thus, placing the e-waste collection boxes around universities can potentially maximise the effectiveness of the said boxes. Fig. 10 illustrates the current household e-waste management framework.

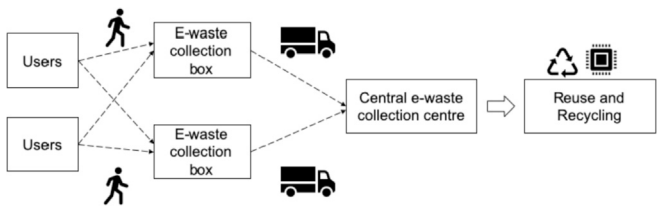


Fig. 10. Illustration of e-waste disposal and recycling process.

The problems faced by the users and recycling company in disposal, collection, and recycling of e-waste are:

- Users will need to know the exact location of the collection boxes to dispose their household waste EEE. However, the collection box locations are often neglected by the residents of the Sunway City (e.g. Kalana, 2010).
- It has been difficult to estimate the flow of waste EEE to the collection boxes (Yong et al., 2019). The company chooses to transfer e-waste from the collection boxes to the central collection centres at fixed intervals because there is no way for the company to predict when the collection boxes will be filled.
- The fixed collection schedule could be counterproductive. If the boxes are full before the scheduled delivery, it might hinder users from continuously disposing their household e-waste.
- The collection boxes are de-coupled from the recycling plant during off-delivery period. There is no way for the e-waste collection company to monitor the collection boxes unless they can remotely monitor the physical boxes. However, it is costly to install closed circuit cameras to remotely monitor them.
- When multiple collection boxes are involved, the e-waste collection schedule is hard to be optimised. In reality, the level of e-waste collection boxes at multiple locations may be different. For example, the collection box at Sunway University might be 90% full while the box at Monash University might be only 10%. Thus, having a fixed collection schedule for all the e-waste collection boxes in the Sunway City area may not be appropriate.

Since the Sunway Group encourages sustainable development and highlights its commitment to achieve sustainable development goals (SDGs) proposed by the United Nations (Sunway Group, 2019), the current work could be used to integrate household e-waste collection boxes in the Sunway City and it could provide innovative e-waste management systems generating maximum benefits to all stakeholders. More importantly, it would foster the SDGs (The United Nations: Sustainable development goals, 2018). In the wider context, it is supposed that the proposed solution could be applicable to manage the household e-waste in the entire country, though a formal collaboration is required, for example with the DOE, Malaysia.

6. Limitations and future works

This work only proposes a cost-effective and user-friendly smart household e-waste management system and thus a few limitations were identified, including possible improvements to be implemented in future studies. Firstly, it is observed that level measurement by the ultrasonic sensor can be unreliable (point source measurement) in certain situations, especially when there are signal interferences. Another reason for the unreliability is because the chosen ultrasonic sensor, HC-SR04, is a low cost sensor and might register false echo when e-waste is deposited into the collection box. To minimise the inaccuracy, multiple sensors or different combination of sensors are suggested. However, the placement of multiple sensors should be carefully tested to avoid the detection of overlapped areas. In addition, the mounting of sensors is limited by the distance from the collection box wall. Recalibration of the sensors is required for different box sizes and sensor geometries. Arduino boards were employed due to its user-friendly nature, though more efficient microcontrollers are recommended in future studies when complex tasks are to be performed. A stable power source is also recommended.

Economic feasibility studies and user surveys should be conducted to explore the economic feasibility (i.e. additional costs per collection box compared to the savings made by the collectors) and

the potential improvements (e.g. mobile apps, reward system) of the proposed household e-waste management system, respectively.

7. Conclusions

The current work is closely aligned with the United Nation's SDGs, such as sustainable cities and communities (Goal 11), and responsible production and consumption (Goal 12) and these are also considered to be the cleaner production aspects of the work. Effective waste management aspects are beneficial to build sustainable cities. Since e-waste generation is one of the fastest in solid waste management in conjunction with mineral resources scarcity and their increased consumption rates, efficient e-waste collection has been critical in urban environments. Even though some countries (e.g. Japan) employ sustainable household e-waste management techniques, including EPR schemes, those have not been fully implemented in developing countries. This work thus addressed the applicability of smart systems to manage the household e-waste pertaining to the Malaysian e-waste recycling sector. A smart e-waste collection box was designed and the users were directed to it using a mobile application. The sensor system and cloud database record the household e-waste disposal data to notify the collector upon the volume change of the e-waste collection box.

Since household e-waste disposal has been challenging in the Malaysian context due to social and legislative constraints, the developed system may be an effective solution to manage household e-waste. The EPR schemes can also be embedded in the developed smart system to encourage the residents to dispose their household waste consumer electronic equipment rather than hoarding of waste EEE. The improvements for the mobile application and the data collection system, and IoT frameworks to connect multiple collection boxes may increase the effectiveness of the developed smart system. The authors also identified the requirements of surveys (e.g. detailed cost analysis, consumer feedback and e-waste collector needs) as future research areas to develop smart household e-waste collection systems for sustainable cities in Malaysia (e.g. Sunway City).

Author contribution section

Kai Dean Kang: Mobile app development, Writing- Original draft preparation, experimental data collection. **Harnyi Kang:** Mobile app development, Writing- Original draft preparation, experimental data collection. **I.M.S.K. Ilankoon:** Conceptualization, Writing- Reviewing and Editing, Supervision. **Chun Yong Chong:** Mobile app validation, Writing- Reviewing and Editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study is supported by the Fundamental Research Grant Scheme, Ministry of Education Malaysia (FRGS/1/2018/TK02/MUSM/03/1). The corresponding author is grateful to the research funding (Grant approval code: GA-SD-17-L01) provided by the GA21 (Global Asia in the 21st Century, Monash University Malaysia) platform. Yun Siew Yong is recognised for her support for the design of the e-waste collection box.

References

Al-Ali, A.R., Zualkernan, I.A., Rashid, M., Gupta, R., Alikarar, M., 2017. A smart home energy management system using IoT and big data analytics approach. IEEE Trans. Consum. Electron. 63, 426–434.
Alcayaga, A., Wiener, M., Hansen, E., 2019. Towards a framework of smart-circular systems: an integrative literature review. J. Clean. Prod. 221, 622–634.
Ari, V., 2016. A review of technology of metal recovery from electronic waste. In: Mihai, F.-C. (Ed.), E-waste in Transition - from Pollution to Resource. IntechOpen, pp. 121–157. https://www.intechopen.com/books/e-waste-in-transition-from-pollution-to-resource/a-review-of-technology-of-metal-recovery-from-electronic-waste.
Baldé, C.P., Forti, V., Gray, V., Kuehr, R., Stegmann, P., 2017. The Global E-Waste Monitor – 2017. United Nations University (UNU), International Telecommunication Union (ITU) and International Solid Waste Association, Bonn/Geneva/Vienna.
Catarinucci, L., de Donno, D., Mainetti, L., Palano, L., Patrono, L., Stefanizzi, M.L., Tarricone, L., 2015. An IoT-aware architecture for smart healthcare systems. IEE. Internet. Things. j. 2, 515–526.
Chen, X., Li, C., Tang, Y., Xiao, Q., 2018. An Internet of Things based energy efficiency monitoring and management system for machining workshop. J. Clean. Prod. 199, 957–968.
Cobbing, M., 2008. Toxic Tech: Not in Our Backyard, Uncovering the Hidden Flows of E-Waste. Greenpeace International, Amsterdam. February 2008.
DOE, 2005. Environmental Quality (Scheduled Wastes) Regulations 2005. Percetakan Nasional Malaysia Berhad - LAWNET. https://www.doe.gov.my/portalv1/wp-content/uploads/2015/01/Environmental_Quality_Scheduled_Wastes_Regulations_2005_-_P.U.A_294-2005.pdf. (Accessed 8 October 2018).
eSWIS, 2019. Electronic Scheduled Waste Information System. https://eswis.doe.gov.my. (Accessed 6 August 2019).
Fernando, W.A.M., Ilankoon, I.M.S.K., Gobal Krishnan, S., Chokshi, V., 2019. The effects of packing shape and structure on liquid distribution of heap leaching systems: addition of PCBs as non-ore particles. Hydrometallurgy 187, 149–157.
Greenpeace International, 2009. Where Does E-Waste End up? accessed. http://www.greenpeace.org/international/en/campaigns/detox/electronics/the-e-waste-problem/where-does-e-waste-end-up/. (Accessed 1 August 2019).
Guo, J., Guo, J., Xu, Z., 2009. Recycling of non-metallic fractions from waste printed circuit boards: a review. J. Hazard Mater. 168, 567–590.
Huang, K., Guo, J., Xu, Z., 2009. Recycling of waste printed circuit boards: a review of current technologies and treatment status in China. J. Hazard Mater. 164, 399–408.
Ilankoon, I.M.S.K., Ghorbani, Y., Chong, M.N., Herath, G., Moyo, T., Petersen, J., 2018. E-waste in the international context – a review of trade flows, regulations, hazards, waste management strategies and technologies for value recovery. Waste Manag. 82, 258–275.
Jayaraman, K., Vejayon, S., Raman, S., Mostafiz, I., 2019. The proposed e-waste management model from the conviction of individual laptop disposal practices – an empirical study in Malaysia. J. Clean. Prod. 208, 688–696.
JICA, 2014. Data Collection Survey on E-Waste Management in Malaysia and Surrounding Countries - Final Report. http://open_jicareport.jica.go.jp/pdf/12154589.pdf. (Accessed 25 August 2018).
Kalana, J.A., 2010. Electrical and electronic waste management practice by households in Shah Alam, Selangor, Malaysia. Int. J. Environ. Sci. 1, 132–144.
Kim, B., Lee, J., Seo, S., Park, Y., Sohn, H., 2004. A process for extracting precious metals from spent printed circuit boards and automobile catalysts. J. Miner. Met. Mater. Soc. 56, 55–58.
Kumar, M., Lee, J.C., Kim, M.S., Jeong, J., Yoo, K., 2014. Leaching of metals from waste printed circuit boards (PCBs) using sulfuric and nitric acids. Environ. Eng. Manag. J. 13, 2601–2607.
Liang, G., Mo, Y., Zhou, Q., 2010. Novel strategies of bioleaching metals from printed circuit boards (pcbs) in mixed cultivation of two acidophiles. Enzym. Microb. Technol. 47, 322–326.
Liu, Y., Zhang, Y., Ren, S., Yang, M., Wang, Y., Huisingh, D., 2019. How can smart technologies contribute to sustainable product lifecycle management? J. Clean. Prod. https://doi.org/10.1016/j.jclepro.2019.119423.
MCMC, 2019. Smart City Initiatives accessed. https://www.mcmc.gov.my/skmmgovmy/media/General/pdf/Sesi-5-Rangkakerja-Berteraskan-Aspek-Informasi-(PLANMalaysia).pdf. (Accessed 6 November 2019).
Meriahtek, 2018. A CSR & Responsible E-Waste Recycling Program Initiated by Meriahtek (M) Sdn. Bhd accessed. http://meriahtek.com.my/ewaste-centre.php. (Accessed 8 October 2018).
Minoli, D., Sohraby, K., Occhiogrosso, B., 2017. IoT considerations, requirements, and architectures for smart buildings-Energy optimization and next-generation building management systems. IEEE. Internet. Things. J. 4, 269–283.
Nižetić, S., Djilali, N., Papadopoulos, A., Rodrigues, J.J.P.C., 2019. Smart technologies for promotion of energy efficiency, utilization of sustainable resources and waste management. J. Clean. Prod. 231, 565–591.
Nowakowski, P., 2018. A novel, cost efficient identification method for disassembly planning of waste electrical and electronic equipment. J. Clean. Prod. 172, 2695–2707.
Omar, M.F., Termizi, A.A.A., Zainal, D., Wahap, N.A., Ismail, N.M., Ahmad, N., 2016. Implementation of spatial smart waste management system in Malaysia. Environ. Earth Sci. 37, 1–9.
Papalambrou, A., Karadimas, D., Gialelis, J., Voyiatzis, A.G., 2015. A versatile scalable

smart waste-bin system based on resource-limited embedded devices. In: 2015 IEEE 20th Conference on Emerging Technologies & Factory Automation (ETFA 2015), September 2015, Luxembourg, pp. 1–8.
Reuter, M.A., van Schaik, A., 2016. Chapter 53: gold – a key enabler of a circular economy: recycling of waste electric and electronic equipment. In: Adams, M.D. (Ed.), Gold Ore Processing. Elsevier, pp. 937–958.
Samann, F.E.F., 2017. The design and implementation of smart trash bin. Acad. J. Nawroz. Univ. 6, 141–148.
Satyamanikanta, S.D., Narayanan, M., 2017. Smart garbage monitoring system using sensors with RFID over internet of things. J. Adv. Res. Dyn. Control Syst. 9, 133–140.
Sepúlveda, A., Schluep, M., Renaud, F., G., Streicher, M., Kuehr, R., Hageluken, C., Gerecke, A.C., 2010. Environ. Impact Assess. Rev. 30, 28–41.
ESP8266 Shop, 2019. ESP8266 Sleep/Low Power Modes. https://esp8266-shop.com/esp8266-guide/sleep-low-power-modes/. (Accessed 6 August 2019).
Sodhro, A.H., Pirbhulal, S., Luo, Z., de Albuquerque, V.H.C., 2019. Towards an optimal resource management for IoT based green and sustainable smart cities. J. Clean. Prod. 220, 1167–1179.
Song, Q., Li, Q., 2015. A review on human health consequences of metals exposure to E-waste in China. Environ. Pollut. 196, 450–461.
Spalvins, E., Dubey, B., Townsend, T., 2008. Impact of electronic waste disposal on lead concentrations in landfill leachate. Environ. Sci. Technol. 42, 7452–7458.
Sunway Group, 2019. Malaysia Takes Leading Role on Sustainable Development accessed. https://www.sunway.com.my/malaysia-takes-leading-role-sustainable-development/. (Accessed 6 August 2019).
Tenaga Nasional Berhad, 2019. Pricing & Tariffs accessed. https://www.tnb.com.my/residential/pricing-tariffs. (Accessed 6 November 2019).
The United Nations: Sustainable development goals, 2018. About the sustainable development goals. https://www.un.org/sustainabledevelopment/sustainable-development-goals/. (Accessed 7 August 2019).
UNEP, 2007a. E-waste Volume I: Inventory Assessment Manual. United Nations Environment Protection.
UNEP, 2007b. E-waste Volume II: E-Waste Management Manual. United Nations Environment Protection.
Van Eygen, E., De Meester, S., Tran, H.P., Dewulf, J., 2016. Resource savings by urban mining: the case of desktop and laptop computers in Belgium. Resour. Conserv. Recycl. 107, 53–64.
Widmer, R., Oswald-Krapf, H., Sinha-Khetriwal, D., Schnellmann, M., Böni, H., 2005. Global perspectives on e-waste. Environ. Impact Assess. Rev. 25, 436–458.
Yong, Y.S., Lim, Y.A., Ilankoon, I.M.S.K., 2019. An analysis of electronic waste management strategies and recycling operations in Malaysia: challenges and future prospects. J. Clean. Prod. 224, 151–166.
Yusof, N.M., Zulkifli, M.F., Yusof, N.Y.A.M., Azman, A.A., 2018. Smart waste bin with real-time monitoring system. Int. J. Eng. Technol. 7, 725–729.